

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element D Initial Template

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a design matrix. Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document. Generate a design blueprint with enough details that someone else not familiar with your project could also build your design.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Design Idea #1

Name: _____

Date: _____

Class: _____

Date: April 13, 2026

Description: A compact, foldable study desk designed for small spaces. It attaches to a wall and can be folded up when not in use. The goal is to save space while still providing a sturdy work surface.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Drawing (description)

Width: 30 inches

- Depth: 18 inches
- Material: Plywood top with metal hinges
- Wall-mounted brackets
- Folding leg support underneath

Design Idea #2

Date: April 13, 2026

Description: A rolling desk with adjustable height. This design allows users to move the desk around easily and change between sitting and standing positions.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Drawing (description):

- Height range: 28–45 inches
- Width: 36 inches
- Materials: wood top, steel frame
- Wheels with locks
- Adjustable telescoping legs

Design Idea #3

Date: April 13, 2026

Description: A multi-functional desk with built-in storage compartments and charging ports. Designed for organization and tech use.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Drawing (description):

Name: _____

Date: _____

Class: _____

- Width: 40 inches
- Depth: 20 inches
- Materials: MDF wood, plastic cable ports
- Built-in drawers (2)
- USB charging hub

Design Idea Comparison and Selection (D1, D2)

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

Criteria + Weights (based on importance):

- Cost (5)
- Durability (4)
- Space Efficiency (5)
- Ease of Build (3)
- Functionality (4)

Score Calculation (Weight × Rating):

- **Option 1:** $(5 \times 4) + (4 \times 3) + (5 \times 5) + (3 \times 4) + (4 \times 3) = 79$
- **Option 2:** $(5 \times 2) + (4 \times 5) + (5 \times 3) + (3 \times 2) + (4 \times 5) = 71$
- **Option 3:** $(5 \times 3) + (4 \times 4) + (5 \times 2) + (3 \times 3) + (4 \times 5) = 70$

Selected Design: Option 1 (Foldable Wall Desk)

Reason: Best balance of space efficiency, cost, and ease of construction.

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	weight	Option 1	Option 2	Option 3	
Cost	5	4	2	2	
Durability	4	3	5	4	
Space Efficiency	5	5	3	2	
Ease of Build	3	4	2	3	
Functionality	4	3	5	5	
score		79	71	70	

Design Blueprint (D4) Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date: April 13, 2026

Sketch:

Manufacturing Methods:

Step 1: Cut plywood to 30" × 18"

- **Step 2:** Attach hinges to back edge of desk
- **Step 3:** Mount desk to wall studs using brackets
- **Step 4:** Install folding support legs
- **Step 5:** Sand and finish surface (paint or varnish)

Supply List and Costs:

Name: _____

Date: _____

Class: _____

Item	Cost each	# needed	Total
Plywood band	20\$	1\$	20\$
Metal Hinges	8\$	2\$	16\$
Wall Brackets	10\$	2\$	20\$
Screws	5\$	1 pack	5\$
Folding Legs	15\$	1 set	15\$
Total Cost	76\$		

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

- Day 1: Research + brainstorming ideas
- Day 2: Sketch designs + choose final design
- Day 3: Gather materials
- Day 4: Cut and assemble parts
- Day 5: Install and test prototype
- Day 6: Evaluate and improve design

Detailed Task Example:

- Measure wood precisely using ruler
- Use saw to cut material



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- Drill holes for screws
- Test folding mechanism for stability

